The documentation and process conversion measures necessary to comply with this amendment shall be completed by 27 July 1991

INCH-POUND

MIL-S-19500/566A
AMENDMENT 3
27 March 1991
SUPERSEDING
AMENDMENT 2
29 June 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON LOGIC-LEVEL, TYPES 2N6902 AND 2N6904, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/566A, dated 8 September 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3, I_{D2} column for 2N6902: Delete "7.6" and substitute "9.6".

1.3, footnote 2/, last line: Delete "(T-25)" and substitute "(T_C-25)".

PAGE 5

4.3.1, second line: Delete "+140°C -0°C," and substitute "+140°C -5°C,".

4.5.2, last line: Delete "$R_{\theta JC}$ (max) = .6°C/W" and substitute "$R_{\theta JC}$ (max) = 1.67°C/W".

4.5.3, delete the main text only (not parameter measurements "a" through "f") and substitute the following:

"4.5.3 Thermal response (ΔV_{SD} measurements). The ΔV_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The ΔV_{SD} conditions (I_H and V_H) and maximum ΔV_{SD} limit shall be derived by each vendor from the thermal response curves (see figure 2). The chosen ΔV_{SD} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report. The chosen ΔV_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots."

---

* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	9
10	10

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:28Z
Check the source to verify that this is the current version before use.

PAGE 7

TABLE I, subgroup 2, gate current, conditions column: Delete "V_{GS} = ±10 V dc" and substitute "V_{GS} = +10 V dc and -10 V dc".

TABLE I, subgroup 2, static drain to source "on-state" resistance, conditions column for 2N6902: Delete "I_D = 7.6 A dc" and "I_D = 5.1 A dc" and substitute "I_{D2} = 9.6 A dc" and "I_{D2} = 5.1 A dc".

TABLE I, subgroup 2, forward transconductance, conditions column: Delete "Pulsed (see 4.5.1)" and substitute "I_D = Rated I_{D2} (see 1.3); Pulsed (see 4.5.1)".

PAGE 8

TABLE I, subgroup 3, gate current, conditions column: Delete "V_{GS} = ±10 V dc" and substitute "V_{GS} = +10 V dc and -10 V dc".

TABLE I, subgroup 3, static drain to source "on-state" resistance, conditions column for 2N6902: Delete "I_D = 7.6 A dc" and "I_D = 5.1 A dc" and substitute "I_{D2} = 9.6 A dc" and "I_{D2} = 5.1 A dc".

TABLE I, subgroup 4, delete in its entirety and substitute the following:

Inspection	Method	Conditions	LTPD 1/	Symbol	Min	Max	Unit
Subgroup 4							
Switching time	3472	I_D = Rated I_{D2} (see 1.3) V_{GS} = 5 V dc Gate drive impedance = 7.5Ω					
Turn-on delay time				$t_{d(on)}$			
2N6902		V_{DD} = 50 V dc			---	50	ns
2N6904		V_{DD} = 100 V dc			---	45	ns
Rise time				t_r			
2N6902		V_{DD} = 50 V dc			---	150	ns
2N6904		V_{DD} = 100 V dc			---	150	ns
Turn-off delay time				$t_{d(off)}$			
2N6902		V_{DD} = 50 V dc			---	130	ns
2N6904		V_{DD} = 100 V dc			---	135	ns
Fall time				t_f			
2N6902		V_{DD} = 50 V dc			---	150	ns
2N6904		V_{DD} = 100 V dc			---	105	ns

PAGE 11

TABLE IIa, subgroup 5, accelerated steady-state operation life, conditions column: Delete "T_J = +200°C" and substitute "(except) T_J = +200°C, +20°C, -0°C". Also in conditions column: Delete "V_{DS} = 20 V dc; P_T = 4 W at".

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:28Z
Check the source to verify that this is the current version before use.

PAGE 12

TABLE IIb, subgroup 3, intermittent operation life (LTPD), conditions column: Delete "A" and substitute "The heating".

PAGE 13

TABLE III, subgroup 2: Delete "Visual and mechanical evaluation, method 2071".

TABLE III, subgroup 6, intermittent operation life (LTPD), conditions column: Delete "A" and substitute "The heating".

PAGE 15

TABLE IV, step 3, gate current, conditions column: Delete "V_{GS} = ±10 V dc" and substitute "V_{GS} = +10 V dc and -10 V dc,".

TABLE IV, step 5, static drain to source "on-state" resistance, conditions column for 2N6902 and 2N6904: Delete "I_D = 7.6 A dc" and "I_D = 5.1 A dc" and substitute "I_{D2} = 9.6 A dc" and "I_{D2} = 5.1 A dc".

TABLE IV, footnote 1/: Delete in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES
NASA - EB13

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13, 15

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1246)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:28Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 5								
Safe operating area test	3474	See figure 3 V_{DS} = 80% of rated V_{DS} and $V_{DS} \leq$ 200 V max						
High voltage test		t_p = 1 s						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7						
Subgroup 6								
Not applicable								
Subgroup 7								
Gate charge	3471	Condition A or B						
On-state gate charge					$Q_{g(on)}$			nC
2N6902						9	19	
2N6904						7	17	
Gate to source charge					Q_{gs}			nC
2N6902						1.1	5.0	
2N6904						1.4	5.0	
Gate to source charge					Q_{gd}			nC
2N6902						5	16	
2N6904						3	14	
Reverse recovery time	3473	$V_{DD} \leq$ 30 V			t_{rr}			
2N6902		di/dt = 100 A/µs I_F = 4 A				---	375	ns
2N6904		di/dt = 100 A/µs I_F = 4 A				---	625	ns

1/ LTPD numbers are to be taken from MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:28Z
Check the source to verify that this is the current version before use.

TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051	Test condition F3, except T_{low} = -55°C		
Hermetic seal a. Fine leak b. Gross leak	1071			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Decap internal visual design verification	2075	See 3.3.2		
SEM	2077			
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately.		
Die shear strength	2017			
Subgroup 4				
Intermittent operation life	1042	Condition D (see 4.3.1), power-on time shall be 3 minutes minimum 2,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle.		
For 2N6902 and 2N6904		T_A = +25°C ±3°C		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8		

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:28Z
Check the source to verify that this is the current version before use.